

AGNIBH DASGUPTA

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Education

2021 - 2026 PhD, Computing & Information Science, **University of Nebraska, Omaha**
2018 - 2020 Master of Science, Computer Science, **Utah State University**
2013 - 2017 Bachelor of Technology, Computer Science, **West Bengal University of Technology**

Publications

- 2026 **TIACam: Text-Anchored Invariant Feature Learning with Auto-Augmentation for Camera-Robust Zero-Watermarking (CVPR)** [🔗](#) (*in press*)
- Combined **CLIP image-text** features with a **learned augmentation module**
 - Exceeded state-of-the-art watermark robustness to camera and environmental distortions
- 2025 **Watermarking Language Models through Language Models (IEEE TAI)** [🔗](#)
- Built a prompt-based **LLM watermarking** framework without modifying model weights
 - Achieved **88%+ watermark detection accuracy** across 25 **open-source LLMs**
- 2024 **Book Chapter, Imaging Science** [🔗](#)
- Improved the scalability of our attention-based watermarking under heavy compound noise
- 2024 **Leveraging Artificial Intelligence (AI) to Enhance CS Instruction (FIE)** [🔗](#)
- Utilized multi-modality in 3D CNN models for mood detection in K-12 classroom videos
- 2023 **Robust Image Watermarking based on Cross-Attention and ID Learning (CSCI)** [🔗](#)
- Designed an attention-based robust image watermarking scheme using **vision transformers**
 - Trained a robust invariant domain using **self-supervised learning**
- 2022 **Perspective transformation layer (CSCI)** [🔗](#)
- Proposed a lightweight layer that learns 2D projections of 3D views for multiple viewpoints

Other Research

- 2026 **Invariant Features in LLMs: Geometric Characterization and Model Attribution**
- Identified semantically meaningful subspace from **9 open-source LLMs**
 - Operationalized these representations for robust zero-shot model attribution
- 2025 **Utilization of SAM2 Model for the Segmentation of Retinal Vascular Leakage on Ultra-wide-field Fluorescein Angiography in Non-infectious Retinal Vasculitis (AAO)**
- Trained a classifier on patches of large optic scans to accurately detect areas of leakage
- 2020 **Deep Q Learning applied to stock trading** [🔗](#)
- Built a **reinforcement learning trading system** using auto-scraped S&P index data

Reviewer Service

2022 - PRESENT ACCV, Scientific Reports - Nature, Springer Nature

Teaching Experience

2022 - 2023 **Intern Supervisor**, University of Nebraska at Omaha
2023 - 2025 **Graduate Teaching Assistant**, University of Nebraska at Omaha
 Courses: Intro to Python, Capstone
2018 - 2020 **Graduate Teaching Assistant**, Utah State University
 Courses: Programming Languages, Intro to C++, Intro to AI

Technical Skills

ML CONCEPTS:	Computer Vision, NLP, Representation Learning, Watermarking, LLMs
FRAMEWORKS:	PyTorch, TensorFlow, HuggingFace, scikit-learn
PROGRAMMING & SYSTEMS:	Python, Bash, Linux, CUDA, Git
DISTRIBUTED & OPTIMIZATION:	DeepSpeed, LoRA, PyTorch DDP, Slurm/HPC